# Title Page

**Manuscript Title:**

Commissioning-Calibrated GP-HOCBF Safety Filtering for Ultra-Supercritical Boiler-Turbine Control under Model Mismatch

**Article Type:**

Original Research Article


**Authors:**

Zhuang Shao[1,*], Lijun Lei[2], Peng Wang[3], Liang Zheng[2], Jie Zhou[2]

**Affiliations:**

1 China Resources Power Technology Research Institute Co., Ltd., Shenzhen 518000, Guangdong Province, China
2 Rundian Energy Science and Technology Co., Ltd., Zhengzhou 450052, Henan Province, China
3 North China University of Water Resources and Electric Power, Zhengzhou 450046, Henan Province, China

**Corresponding Author:**

Zhuang Shao
China Resources Power Technology Research Institute Co., Ltd.
Shenzhen 518000, Guangdong Province, China
Email: shaozhuang@crpower.com.cn

**ORCID:**

Zhuang Shao: 0000-0003-2496-0797



**Acknowledgements:**

Not applicable.

**Funding:**

This work was supported by the Postdoctoral Research Project of China Resources Power Technology Research Institute Co., Ltd. (Internal Project No. 374322).


**Declaration of Conflicting Interests:**

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

**Data Availability Statement:**

The source code, simulation scripts, benchmark results, plotting scripts, derived plant-historian metrics, and bounded controller-export excerpts are available at https://github.com/VickylastShao/Robust-HOCBF-Safety-Filtering-for-Supercritical-Power-Plants-under-Model-Mismatch.
Complete raw plant records are proprietary enterprise assets. Qualified researchers may request restricted access from the corresponding author, subject to data-owner approval and an executed data-use agreement.

**Author Contributions:**

Zhuang Shao conceived the study, developed the methodology, performed the simulations, analyzed the results, and wrote the original draft.
Lijun Lei provided engineering requirements, plant-data interpretation, and field-validation support.
Peng Wang contributed to control-system analysis and manuscript review.
Liang Zheng supported plant-data preparation and validation-log interpretation.
Jie Zhou contributed to deployment-context analysis and manuscript review.
All authors reviewed and approved the final manuscript.

**Number of words:**

9432

**Number of figures:**

8

**Number of tables:**

5